SHUBHASHIS SINHA

Principal Engineer — Generative AI | Enterprise Agentic AI Platforms (Greenfield & Brownfield) · HA & 100x Scale · Agent Eval · AI Observability & Trust

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SUMMARY

Hands-on Principal-level engineer and AI architect with **12+ years** who loves building **enterprise-grade Generative AI platforms and systems from scratch** — greenfield and brownfield — with a focus on **high availability, scaling to 100x, and research on emerging AI technologies**. Deep expertise building **large enterprise-scale agentic AI platforms and services**, **agent evaluation, AI observability, and AI trust / content moderation**. A passionate coder and solution builder: Python-first, distributed-systems background, and production rigor across reliability, latency, and cost. Patent filing in progress for a GraphRAG SPARQL Validator. Currently Staff AI Architect in GE HealthCare's AI Center of Excellence.

CORE COMPETENCIES

Agentic AI Platforms	Enterprise-Scale Agentic AI Platforms & Services · Greenfield & Brownfield Builds from Scratch · Multi-Agent Orchestration · Tool Use · Agent Memory · LangGraph · AWS Bedrock AgentCore · MCP
HA, Scale & Performance	High Availability · Scaling to 100x · Distributed Systems · Microservices · Async / Streaming · Caching · Model Routing · Latency / Cost / Resource Optimization · Resilience by Design
Agent Eval & Observability	Agent Evaluation (accuracy, task success, groundedness, latency, cost) · Evaluation Harnesses · RAGAS · LLM-as-Judge · AI Observability · Tracing / Telemetry · Monitoring at Scale
AI Trust & Safety	AI Trust · Content Moderation · Guardrails · Hallucination Mitigation · Prompt-Injection Defense · Responsible AI · Data Privacy · Auditability
GenAI & Retrieval	LLM Applications · RAG / GraphRAG · Vector Databases · Embeddings · Semantic Search · Knowledge Graphs (Amazon Neptune) · Speech AI (framework-aware)
Engineering & Cloud	Python (primary) · SQL · APIs · Event-Driven Systems · AWS (Bedrock, EKS, Lambda, S3, OpenSearch) · Docker / Kubernetes · CI/CD · Research into Emerging AI · Mentoring

PROFESSIONAL EXPERIENCE

Staff AI Architect — AI Center of Excellence

Dec 2021 – Present

GE HealthCare · Bangalore, India

Enterprise Agentic AI Platforms — Build from Scratch, HA & Scale

- Built **enterprise-scale agentic AI platforms and services from scratch** (greenfield) and evolved existing systems (brownfield) — multi-agent orchestration, tool use, and memory (LangGraph, AWS Bedrock AgentCore, MCP).
- Engineered for **high availability and large-scale performance** — distributed, microservice, event-driven architecture with async/streaming, caching, and model routing (enterprise RAG over 10,000+ documents at sub-second P95).
- Optimized across **latency, cost, and resource usage** as query volume grew.

Agent Evaluation, AI Observability & Trust

- Built **agent evaluation** — accuracy, task success, groundedness, latency, cost — with evaluation harnesses (RAGAS, LLM-as-Judge); patent filing in progress for a GraphRAG SPARQL Validator (~95% correctness).
- Designed **AI observability** — tracing, telemetry, and monitoring of agent/LLM behavior at scale.
- Built **AI trust and safety** — guardrails, hallucination mitigation, prompt-injection defense, content moderation, and auditability.

Research, Reusable Platforms & Leadership

- Researched and adopted **emerging AI technologies**, translating them into production platform capabilities; GraphRAG on Amazon Neptune (131 predicates / 3 namespaces).
- Built reusable platform capabilities adopted by **6+ teams** (25% faster; ~40% effort reduction across 4+ teams / 20+ developers).
- **Mentored 8+ engineers** as a hands-on builder; delivered a platform migration ($2M+ savings).

Tech Stack: Python · Agentic AI · LangGraph · AWS Bedrock AgentCore · MCP · RAG / GraphRAG · Agent Eval (RAGAS, LLM-as-Judge) · AI Observability (tracing/telemetry) · Guardrails / Content Moderation · Vector DBs · AWS (Bedrock, EKS, Lambda, S3, OpenSearch) · Docker / Kubernetes · Async/Streaming · CI/CD

Senior Software Engineer — Distributed Systems & Platform

Oct 2017 – Dec 2021

Bank of America · Chennai, India

- Built **large-scale, highly available distributed platforms and services** for a global bank — **5TB+ daily** — engineered for reliability and scale.
- Built APIs, microservices, and event-driven systems with observability, security, and automated validation.
- Optimized performance 40% on 100B+ row datasets; reduced manual QA 30%; mentored engineers.

Tech Stack: Python, SQL, Apache Spark, Kafka, AWS (S3, EMR, Glue), REST APIs, microservices, event-driven, CI/CD

Software / Data Engineer

Mar 2014 – Oct 2017

T-Mobile · Bangalore, India

- Built **large-scale distributed systems and pipelines** (Hadoop, Pig, Hive) across **5M+ subscribers** — 25% improvement in data-driven decisions.
- Implemented **CDC/incremental ingestion** and tuned query performance 30%.

Tech Stack: Hadoop (CDH), Apache Pig, Hive, Sqoop, Python, SQL

EDUCATION

Bachelor of Technology — Electronics & Communications Engineering

West Bengal University of Technology · Kolkata, India · GPA: 8.87/10

AWARDS & CERTIFICATIONS

🏆 **Sir M. Visvesvaraya Award — GE HealthCare (2024)**

Recognized for enterprise agentic AI platforms, evaluation, and observability with measurable production impact.

🏆 **Impact Award — GE HealthCare (Oct 2022, Nov 2023, Mar 2024)**

Reusable AI platform capabilities, platform migration delivering $2M+ savings, and AI-driven automation.

☁️ **AWS Certified Solutions Architect – Associate (2025) · Patent filing in progress (GraphRAG SPARQL Validator architecture)**